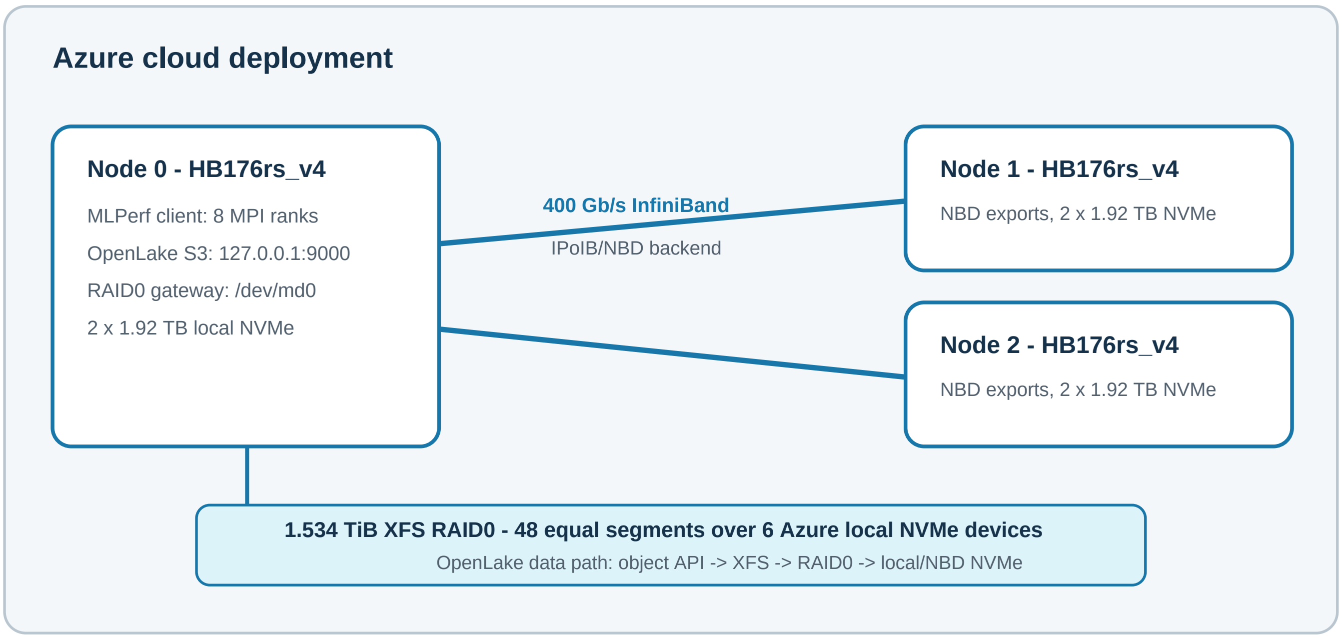
OpenLake HBv4 Object Storage

System name: openlake-hbv4-1c1s-final

Field	Submission value
Submitter	OpenLake
Submission date	2026-07-24
Benchmark	MLPerf Storage v3.0 - Checkpointing, Llama3-8B
Division	CLOSED
Solution type	Cloud-deployed S3-compatible object storage
Deployment	Three Azure Standard_HB176rs_v4 virtual machines
Power requirements	Cloud deployment; customer-provisioned power table is not applicable

System topology



The MLPerf checkpointing load generator and the OpenLake S3 gateway ran on Node 0. Node 0 striped storage across 16 local loop-backed segments and 32 NBD segments exported by Nodes 1 and 2 over the Azure InfiniBand fabric. The benchmark accessed OpenLake through the S3 object API on loopback; the storage backend used IPoIB/NBD.

Hardware and software configuration

Component	Quantity and configuration
Client / gateway	1 x Azure Standard_HB176rs_v4; 176 CPU cores; 756 GiB RAM; 2 x 1.92 TB Azure NVMe Direct Disk
Backend nodes	2 x Azure Standard_HB176rs_v4; each with 176 CPU cores, 756 GiB RAM, and 2 x 1.92 TB Azure NVMe Direct Disk
Data network	Azure 400 Gb/s InfiniBand fabric used for IPoIB/NBD backend traffic
Storage layout	1.534 TiB XFS filesystem on Linux md RAID0; 48 x approximately 32 GiB segments distributed across all six NVMe devices
Operating system	Ubuntu 22.04 on all three nodes
Benchmark client	MLPerf Storage 3.0.46 at Git commit 1a236dc3e00b0578a8c03d7ed81b3f54d0329637; DLIO 3.0.4; s3dlio 0.9.112
Storage server	OpenLake optimized build; binary SHA-256 bbe069bca96d188fdbfcc2d5766bff89dc714c00fd93a0a7bf4896bdc0370b87

Runtime configuration summary

Area	Configuration used for the measured run
Benchmark client	Unmodified MLPerf Storage client and locked dependencies using the S3 object API.
OpenLake data path	OpenLake S3 service on XFS RAID0 spanning six Azure local NVMe devices.
Data layout	Large streaming data/read windows with parity disabled for the measured single-logical-drive configuration.
Host tuning	File-descriptor and Linux writeback limits sized for concurrent checkpoint I/O.